The Multi-Modal Frontier: Synergizing Proteomics and Imaging for Precision Diagnostics and Mechanistic Discovery in Alzheimer's Disease

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***Abstract*—Biological aging varies substantially across individuals and is a major contributor to Alzheimer's disease risk and progression. While neuroimaging-based "brain age" models capture macrostructural changes in brain anatomy, circulating proteomic profiles reflect systemic processes, such as inflammation, vascular dysfunction, and metabolic stress that also shape neurodegeneration. Here, we present a multimodal classification framework that integrates proteomic biomarkers with structural magnetic resonance imaging (MRI) features. Using regularized machine learning with rigorous cross-validation, we demonstrate that multimodal fusion improved classification performance relative to either clinical, proteomics or MRI alone. The multimodal model achieved macro-F1 = 0.77, recall = 0.74, precision = 0.79 and accuracy = 0.84, compared with macro-F1 = 0.46 using demographic features, 0.60 for proteomics and 0.56 for MRI features alone, respectively.**

***Index Terms*—Cross-Attention Fusion, Neurodegeneration, Biological Age, MRI, Proteomics.**

I. Introduction

Alzheimer's disease (AD), the most common cause of dementia, is a progressive brain disorder that slowly destroys memory and thinking skills, resulting in confusion and disorientation, language difficulty, issues with judgement and decision making, and changes to mood and personality, culminating in loss of independence [1]. Due to its global impact, it has been designated by the World Health Organization (WHO) as a major public health challenge. In the United States alone, over 7 million Americans 65 years of age and older (approximately 1 in 9) are living with AD, expected to double by 2060 [2]. The clinical presentation of AD includes memory loss, particularly for recent events as a result of neuronal cell death in sensitive brain regions such as the entorhinal cortex [3]. Strikingly, the pathophysiology underlying symptoms begins decades prior, with multifactorial etiology due to genetics, environmental factors and exposures [1]. Thus, reliance on clinical symptoms limits the ability to predict and address AD [4]. The hallmark pathological features of AD – beta-amyloid plaques, tau tangles, and neuroinflammation – have been well-defined in post-mortem tissue [1]. Recently, biomarkers such as beta-amyloid and tau proteins in cerebrospinal fluid (CSF), positron emission tomography (PET) and magnetic resonance imaging (MRI) have become established [3, 4]. Such biomarkers may show evidence of disease even before cognitive and clinical symptoms [5]. Standardized, reliable and sensitive blood-based biomarkers, once established, would enable lower cost and less invasive methods for prediction and progression monitoring across the AD continuum [3]. While a simplification, this AD continuum has been broadly categorized largely based on cognitive testing in conjunction with the biomarkers and clinical presentation into AD, cognitive normal (NL) and mild cognitive impairment (MCI) patients. This classification system provides a framework suitable for application of machine learning approaches.

Most machine learning models have employed either imaging modalities (such as MRI and PET) with clinical tests or molecular data (such as CSF biomarkers) independently as features for prediction [6, 7]. Because these different features provide orthogonal information, there could be benefits in combining them [3, 8–14]. However, using only established biomarkers limits the ability for early detection and intervention. Newer sources, such as plasma proteomic studies, provide opportunities for the potential discovery of novel drug targets and biomarkers of disease risks [15, 16]. Proteomic datasets vary substantially in size and platform, ranging from single protein datasets with high specificity, to multiple proteins, sometimes in the thousands [15]. The urgent task is to extract novel blood-based biomarkers from these large-scale proteomics studies that could be integrated with the established AD biomarkers. Traditionally, this has been done by NL/MCI/AD classification using the proteomics markers alone [17–19], but potential benefits of an earlier integration of imaging with proteomics are emerging [20, 21], e.g. for voxel-wise regression of individual proteins against imaging features to predict MCI-to-AD conversion [22], and for unsupervised subtyping of MCI rather than supervised diagnostic classification [23]. However, these efforts have largely relied on fixed concatenation or late-stage fusion of modalities, have predominantly used CSF- or PET-derived molecular features rather than plasma proteomics, and have not jointly modeled diagnostic classification together with biological-age regression within a single end-to-end framework. The synergy between neuroimaging and proteomics, when coupled with adaptive fusion and multitask learning, may enable next-generation personalized diagnostics and targeted therapies.

In this paper, we developed a multimodal approach that integrates plasma proteomic features with structural MRI through cross-attention fusion, jointly performing NL/MCI/AD classification and brain-age regression (Fig. 1), which outperforms all single feature type models constructed for comparison. The fusion model outperforms all single-feature-type models constructed for comparison and demonstrates a more balanced performance across all three classes. Below, we provide the details of how the models were constructed and evaluated, followed by receiver operator characteristics and confusion matrix analyses.

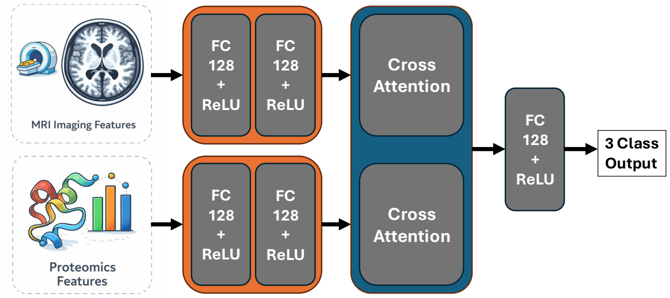

Fig. 1. Pipeline of proposed muti-modal classification system. Both MRI imaging features and proteomics-based features are passed through fully connected (FC) embedding layers with rectified linear unit (ReLU) activations and combined via cross-attention. A 128-dimensional output head is used to perform classification.

II. Methods

The classification component of the proposed framework was designed to distinguish among three diagnostic groups: NL, MCI and AD. The baseline cohort used for classification consisted of 500 patients in total, including 49 NL, 356 MCI, and 95 AD.

A. ADNI Data Extraction and Demographics

Data for this study were obtained from the ADNI database. We began with the ADNI plasma proteomic dataset and limited the analysis to baseline samples. To construct the multimodal cohort, we then identified structural MRI data corresponding to the same subjects. After this matching procedure, the final study cohort consisted of 500 patients, with 49 NL, 356 MCI, and 95 AD subjects, respectively. The distributions of the male and female participants in NL, MCI and AD were 49% and 51%, 63% and 37%, and 58% and 42%, respectively. NL, MCI, and AD cohort ages ranged from 62 - 89.6 years, 54.4 - 89.3 years, and 55.1 - 89.1 years, with mean ages of 74.9 $\pm$ 5.8, 74.7 $\pm$ 7.5, and 75.3 $\pm$ 8.0 years, respectively. The burden of APOE4 increased with the severity of the diagnosis. Among NL participants, 44 of 49 participants (89.8%) had no APOE4 alleles, 5 (10.2%) had one APOE4 allele, and none had two APOE4 alleles. Among MCI participants, 169 of 356 (47.5%) had no APOE4 alleles, 142 (39.9%) had one APOE4 allele, and 45 (12.6%) had two APOE4 alleles. Among AD participants, 31 of 95 (32.6%) had no APOE4 alleles, 42 (44.2%) had one APOE4 allele, and 22 (23.2%) had two APOE4 alleles. Overall, APOE4 carriers represented 10.2% of the NL group, 52.5% of the MCI group, and 67.4% of the AD group. The cohort was predominantly White across all diagnostic groups. Among NL participants, 45 of 49 were White (91.8%) and 4 were Black (8.2%). Among MCI participants, 334 of 356 were White (93.8%), 12 were Black (3.4%), 9 were Asian (2.5%), and 1 was American Indian or Alaskan (0.3%). All 95 AD participants were White (100.0%). Overall, the cohort included 474 White participants (94.8%), 16 Black participants (3.2%), 9 Asian participants (1.8%), and 1 American Indian or Alaskan participant (0.2%).

Ethnicity data showed that 483 participants were not Hispanic or Latino (96.6%), 13 were Hispanic or Latino (2.6%), and 4 had unknown ethnicity (0.8%). For education levels, NL participants had an average of 15.9 $\pm$ 2.6 years of education, ranging from 10 to 20 years; MCI participants had an average of 15.7 $\pm$ 3.0 years, ranging from 6 to 20 years; and AD participants had an average of 15.2 $\pm$ 3.1 years, ranging from 4 to 20 years. Baseline cognitive and functional measures showed the expected diagnostic gradient, with mean MMSE scores decreasing from 29.0 $\pm$ 1.1 in NL to 27.0 $\pm$ 1.8 in MCI and 23.6 $\pm$ 1.9 in AD. In contrast, CDR-SB scores increased from 0.03 $\pm$ 0.12 in NL to 1.58 $\pm$ 0.88 in MCI and 4.25 $\pm$ 1.51 in AD. ADAS-Cog 13 scores increased from 9.5 $\pm$ 4.4 in NL to 18.6 $\pm$ 6.1 in MCI and 28.2 $\pm$ 7.6 in AD, and FAQ scores increased from 0.04 $\pm$ 0.20 in NL to 3.82 $\pm$ 4.48 in MCI and 12.23 $\pm$ 6.76 in AD.

B. ADNI Proteomics and Data Processing

Plasma proteomics was performed using Rule-Based Medicine (RBM) Human DiscoveryMAP multiplex immunoassay platform implemented on the Luminex xMAP system. 190 biomarkers were measured in EDTA plasma samples, collected following overnight fasting according to ADNI protocols. Samples were randomized prior to assay processing, with laboratory personnel being blinded to clinical diagnosis and participant demographics. Quality-control (QC) samples representing low, medium, and high analyte concentrations were included on each assay to monitor precision and batch variability. Samples were analyzed in singlicate and QC controls were analyzed in duplicate. Biomarkers with more than 10% missing values ("QNS") or more than 10% measurements below the lower limit of detection ("LOW") were excluded. Remaining "LOW" values were imputed as one-half the lower limit of detection. Other missing values were imputed using biomarker-specific mean imputation. Samples containing imputed values for greater than 25% of biomarkers were excluded. Empty cells, negative and non-numeric values were imputed to avoid biologically implausible abundance estimates. 44 biomarkers with missing data were removed and outlier measurements exceeding 5 standard deviations from the biomarker mean were replaced with the nearest non-outlier value. Multidimensional scaling with Mahalnobis distance-based approaches were used to identify potential outliers and misclassified subjects. Biomarker distributions were evaluated individually and transformed where necessary using Box-Cox transformations and standardized using z-score normalization (mean-centered and variance-scaled) implemented with scikit-learn StandardScaler. Potential technical effects related to batches, run date, and QC characteristics were evaluated using graphical inspection and plate-level QC monitoring without batch correction.

C. MRI Feature Characteristics

The MRI modality was represented using tissue-derived volumetric characteristics obtained from structural brain images after regional segmentation. In particular, the extracted features described three major tissue compartments: gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF). Gray matter features captured the volume of cortical and subcortical tissue that contains neuronal cell bodies and is commonly associated with regional atrophy in AD. WM features described the volume of myelinated fiber pathways that support communication between brain regions and may reflect structural disconnection or degeneration. CSF features provided an indirect marker of tissue loss, since expansion of CSF spaces often accompanies brain atrophy. From these segmented tissue maps, the model used global volumetric summaries including total GM, total WM, total CSF, and total brain volume, together with the mean and standard deviation of each tissue type across regions and normalized tissue-to-brain volume ratios. In addition to these global measures, the MRI representation also included targeted regional features from AD-sensitive areas, allowing the model to capture both whole-brain structural composition and localized neurodegenerative patterns.

D. Demographic Variables Model

A baseline demographic variable model was trained on the three-class classification task. Age, race, gender, apolipoprotein e4 (APOE4) genotype, marital status, and years of education were used as predictors in a multinomial logistic regression model with L1 penalty.

E. Proteomics Only Model

Multivariate analyses were performed separately for baseline ("bl") and 12-month ("m12") visits using partial least squares discriminant analysis (PLS-DA) using scikit-learn PLSRegression framework. Diagnostic groups consisted of cognitive normal (CN), mild cognitive impairment (MCI), and Alzhiemer's Disease (AD). To address class imbalance across diagnostic categories, balanced datasets were generated using Synthetic Minority Oversampling Technique (SMOTE) paired with bootstrap upsampling with replacement used as a fallback when SMOTE was not available. Variable importance in projection (VIP) scores were calculated from fitted PLS-DA models using weighted sums of squares derived from latent variable loadings and scores. 65 biomarkers with VIP scores greater than one were considered significant contributors to diagnostic discrimination.

F. Multimodal Feature Integration Model

1) Multimodal Architecture: The model used a multimodal neural network to integrate MRI-derived summary features with plasma proteomic features for simultaneous brain-age regression and three-class diagnosis (see Fig. 1). The MRI input was represented by global brain summary measures together with AD-sensitive regional features, while the proteomic input consisted of a fixed VIP-selected biomarker panel derived from prior PLS-DA analysis. Each modality was first processed by its own multilayer perceptron in order to learn modality-specific feature embeddings. The MRI branch learned a compact representation of structural imaging features, and the proteomic branch learned a corresponding embedding of the selected blood-based biomarker profile. These modality-specific embeddings were then combined using a cross-modal attention fusion module, allowing the network to adaptively weight the relative contribution of MRI and proteomic information at the individual-subject level rather than relying on a fixed concatenation scheme. The fused representation was passed through a shared latent encoder, which produced a common feature space used by both downstream tasks.

The CrossModalAttentionFusion is attention-style gated fusion, and its main advantage for this task is per-sample modality weighting. This is advantageous in the present task, because proteomic changes can precede structural atrophy in early disease while late AD shows clear MRI atrophy which will be accompanied by proteomic changes. Thus, the gate can lean on whichever modality is informative for a given subject at a given stage of the disease. That makes it strictly more flexible than averaging, more data-efficient than concatenation+MLP, and able to capture conditional cross-modal interactions that late fusion and independent encoders structurally cannot — plus it produces the single fused embedding your age head, BAD-feedback channel, and center loss all depend on. The caveat is that for two low-dimensional modalities the empirical edge over plain concatenation is often small or zero, and the extra parameters can overfit at small N. Gated fusion is expected to give per-sample weighting with a favorable inductive bias, which we have verified empirically through an ablation study where we used the same folds and seed, while only swapping the fusion block which resulted in loss of performance when only imaging data were used. Finally, the use of a weighted gate makes the model more interpretable and highlights which brain regions of interest and proteomic features are preferentially used together in the prediction providing potential hypotheses for the underlying biological mechanisms.

2) Regression Branch: From the shared latent representation, the model included a regression branch that predicted chronological brain age as a continuous output. The age-regression component was trained using mean squared error loss between predicted age and chronological age. Rather than restricting regression to only one diagnostic group, the regression head was trained using the full dataset so that all subjects contributed to the estimation of age-related structure in the multimodal feature space. To further guide the regression branch toward clinically meaningful progression patterns, the model incorporated a brain-age-delta margin ranking loss. This auxiliary loss encouraged the mean brain-age delta to follow the expected disease ordering, such that lower-severity groups had lower average brain-age delta than higher-severity groups. As a result, the regression objective combined a standard continuous prediction target with an ordinal regularization term designed to preserve biologically plausible separation across NL, MCI, and AD.

3) Classification Branch: The second task-specific branch was a classification head that produced a probability distribution over the three diagnostic classes, enabling final prediction of NL, MCI, or AD. In addition to the shared latent representation, the classification head also received a brain-age-delta feedback signal, defined as the difference between predicted brain age and chronological age. This feedback pathway allowed age-related deviation information to inform diagnostic classification while preventing classification gradients from directly distorting the regression output. Because the diagnostic categories follow a natural disease-ordering progression, the classification objective was formulated to account for both class imbalance and ordinal severity. The primary classification loss was Focal Loss with label smoothing. Focal Loss emphasized harder-to-classify samples and reduced the influence of easily classified cases, which was especially important because the dataset was dominated by the MCI group. Label smoothing was incorporated to reduce overconfident predictions and improve generalization. In addition, an ordinal penalty matrix was applied so that clinically larger mistakes were penalized more heavily than adjacent-stage errors. For example, misclassifying an NL subject as AD, or an AD subject as NL, incurred a stronger penalty than confusing NL with MCI or MCI with AD. To improve class separation in the latent space, the model also incorporated Center Loss as an auxiliary regularization term during training. Center Loss encouraged samples from the same diagnostic group to cluster around a class-specific centroid in the embedding space, thereby promoting a more compact and discriminative latent structure.

4) Training and Preprocessing: The overall multitask objective combined regression loss, classification loss, center-based regularization, and the brain-age-delta ranking loss in a weighted manner. This design allowed the model to learn a shared multimodal representation that supported both continuous age prediction and categorical disease discrimination. In practice, the regression branch encouraged the network to preserve age-related structure, whereas the classification branch emphasized disease separation. The ranking constraint and the brain-age-delta feedback pathway linked the two objectives by explicitly encouraging age-related deviation to align with disease severity.

The original PLS-DA and logistic-regression baselines were intended to represent conventional single-modality practice, but we acknowledge they do not control for classifier strength. To address this, we added input ablations in which the identical multimodal architecture — same gated fusion, multitask heads, focal loss, ordinal penalty, center loss, calibration, oversampling, and early stopping — is trained on each modality alone on the same folds and seed, so that capacity is held constant and only the input modalities vary; the full model outperforming each matched unimodal variant isolates the gain as complementarity rather than a stronger classifier. We complement this with the reverse control, applying strong nonlinear classifiers to each single modality, and a naive-fusion baseline to verify the contribution is multimodal rather than specific to our fusion design. All multimodal-versus-unimodal differences are now reported with paired tests across the shared CV folds and fold-wise variance, so the reported improvements are shown to exceed fold-level noise.

To address the substantial class imbalance in the training data, GNUS-based oversampling was applied within each training fold only. The oversampling strategy used heavier augmentation for the normal group and additional augmentation for the AD group in order to reduce the dominance of the MCI class during optimization. This augmentation was performed only on the training partition of each fold so that the validation data remained untouched and reflected true held-out performance. All MRI features, proteomic features, and age targets were standardized using statistics computed from the training set only, and the same fitted scalers were then applied to the corresponding validation set to prevent information leakage. Because the proteomic modality was defined using a fixed VIP-selected biomarker panel rather than fold-specific feature selection, the proteomic input remained constant across folds, which provided a cleaner comparison of multimodal classification and regression performance.

G. Model evaluations

Classification performances were evaluated using accuracy, macro-averaged F1 score, and confusion matrices, with macro-averaged measures to account for class imbalance. Class-wise discrimination was assessed using one-vs-rest receiver operating characteristic (ROC) curves, with overall performance summarized by area under the curve (AUC). The multimodal model employed 5-fold stratified cross-validation, with optimization via Adam, cosine annealing, gradient clipping, and early stopping based on validation macro-F1. Per-class probability thresholds were calibrated using coordinate descent to improve macro-F1 and reduce majority-class bias. The demographics model followed the same general evaluation framework using ROC curves, macro-F1, and accuracy. The proteomics model used an 80/20 train–test split, with feature scaling fit on training data and SMOTE applied only to the training set. A PLS-DA classifier was trained on VIP-selected biomarkers, and performance was additionally quantified using balanced accuracy, macro-precision, macro-recall, Matthews correlation coefficient, and classification reports.

III. Results

The multitask performance of the proposed models were evaluated using 5-fold cross-validation on three diagnostic classes, NL/MCI/AD. Using demographic features alone to classify the groups (age, APOE4 status, marriage status, years of education, and gender; Figure 2A served as a baseline comparison. These baseline features achieved an accuracy of only 66% and a macro-F1 score of 0.46. This performance was not affected by age, despite the known age-dependence of AD. This is because the ADNI dataset we used was age controlled, balancing the representation of age ranges across all three classes. In the following, we will show that all of the other models substantially outperformed demographic information alone.

Figure 2B shows the evaluation of the performance of classifying the diagnostic groups using only MRI imaging features. The ROC analysis of this model showed uneven class separability across the three diagnostic groups. The model performed best for the NL class, with an AUC of 0.8529, indicating strong discrimination of cognitively normal subjects. In contrast, performance was much weaker for MCI and AD, with AUC values of 0.4162 and 0.5550, respectively. The overall AUC of 0.6238 suggests only modest diagnostic discrimination across all three classes, indicating that the model was most effective at identifying NL, while separation of the disease groups remained limited.

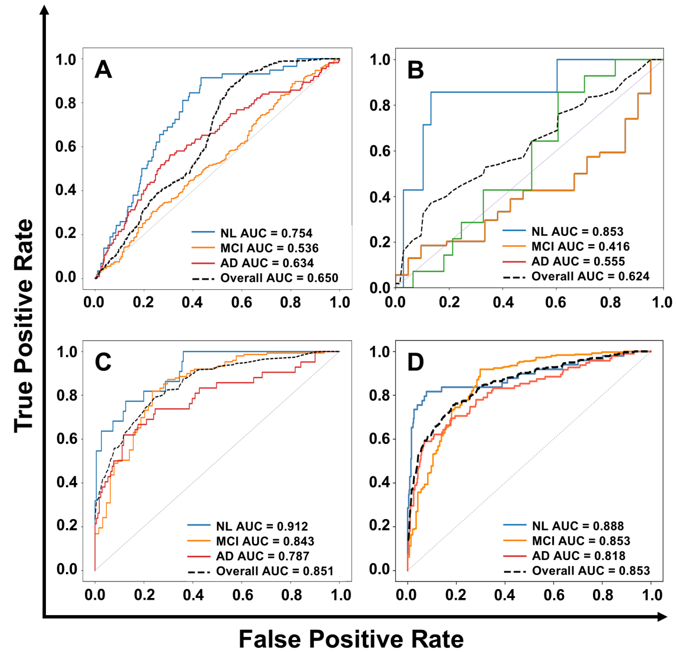

Fig. 2. One-vs-rest receiver operating characteristic (ROC) curves for NL, MCI, and AD classification using: A). Demographics features where the highest area under curves (AUCs) were observed for NL (AUC = 0.758), AD (AUC = 0.56), and MCI (AUC = 0.64). B). MRI imaging features where the highest AUCs were observed for NL (AUC = 0.853), AD (AUC = 0.555), and MCI (AUC = 0.416). C). Proteomics features where the highest AUCs were observed for for NL (AUC = 0.912), MCI (AUC = 0.843), and AD (AUC = 0.787). D). Multi-modal features from MRI imaging and proteomics where the highest AUCs were NL (AUC = 0.888), MCI (AUC = 0.853), and AD (AUC = 0.818).

Figure 2C displays the performance of using only proteomics features to classify the groups. Using a PLS-DA model to determine protein drivers of separation, 65 quantified proteins recorded at baseline achieved an accuracy of 67% with a macro-F1 score of 0.6 as summarized in Table I, slightly higher than the imaging-based features (with an F1 of 0.56). Overall, the multimodal approach (Figure 2D) outperforms the PLS-DA model.

Finally, the fusion model achieved an overall accuracy of 0.840 and a macro-F1 score of 0.77, indicating strong multiclass diagnostic performance, better than any other model using a single type of feature alone. These results indicate that the model was able to capture meaningful variation while also providing strong diagnostic discrimination. The analysis of the

TABLE I
MODEL PERFORMANCE COMPARISON

Model	F1	Recall	Precision
Demographics	0.46	0.43	0.54
MRI Images	0.56	0.67	0.58
Proteomics	0.60	0.67	0.57
Multi Model	0.77	0.74	0.79

class-specific prediction behavior revealed improved precision for NL and AD relative to the earlier feature configuration. However, AD sensitivity remained lower than for the other groups. The stronger MCI recall suggests that the learned multimodal representation was particularly effective at identifying the dominant intermediate disease group. However, the lower AD recall indicates continued overlap between MCI and AD, suggesting that these two groups share structural and proteomic characteristics that are more difficult for the model to separate. Overall, the results demonstrate that the proposed multitask model provides strong three-class diagnostic performance, supporting its usefulness as a joint framework for disease classification and age-related modeling.

IV. DISCUSSION

As increased age is the most significant risk factor for AD, we were initially surprised to see the lack of effect of age as a variable on the baseline classifier. However, since the ADNI dataset explored here is age-balanced across the three classes, it provides us with an opportunity for a more fine-grained analysis of age. This is important in the context of health and disease, because individuals of the same chronological age often exhibit variability in disease risk and survival, referred to as biological age. Many different biomarkers have been explored to capture biological age, such as epigenetic [24] or proteomic markers [25]. Differential rates of proteomic aging across tissues correlate with distinct phenotypes and endpoints [26–28] and may also allow disease risk stratification across APOE haplotypes, underscoring biological resilience in individuals with super-youthful brain signatures [27]. To investigate if this heterogeneity may be captured by the multimodel feature analysis, we evaluated the multitask performance of the fusion through brain-age regression (Fig. 3). We can see that AD patients tend to have older biological ages as compared to their chronological ages, with the reverse for MCI patients. This is reflected quantitatively in the mean absolute error (MAE) of 4.96 $\pm$ 0.49 years, a root mean squared error (RMSE) of 6.31 $\pm$ 0.43 years, an R^2 value of 0.260 $\pm$ 0.139, and a Pearson correlation of 0.551 $\pm$ 0.095 between predicted brain age and chronological age. The relatively low R^2 is a clear indicator that biological age can be substantially lower or higher in individuals. These results indicate that the fusion regression model was able to capture meaningful age-related variation while the classification success highlights that it also provides strong diagnostic discrimination.

Our results show that the use of brain-age-delta feedback to inform classification is successful at linking regression and classification tasks. Furthermore, the use of GNUS based oversampling and SMOTE on the training set to balance NL, AD and MCI groups helps improve performance. Interestingly, our model has the most difficulty in differentiating MCI and AD, while previous classifiers often confuse NL and MCI groups due to overlapping transitional disease features [29]. However, prior work focused on binary classification while differentiating all three classes simultaneously is a harder but also clinically more relevant task as we wish to diagnose a given patient at any stage of the disease. Confusing MCI and AD is less serious than NL and MCI, because early detection is more important as it may still be possible to delay progression of the disease.

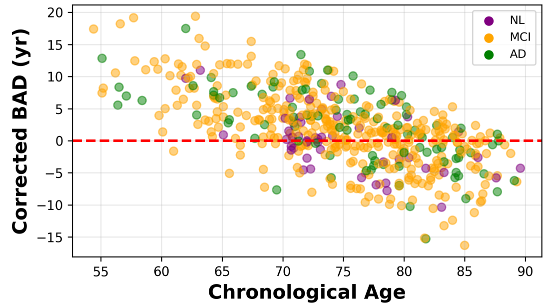

Fig. 3. Relationship between corrected brain-age delta (BAD) and chronological age across participants in the NL, MCI, and AD groups. Each point represents one subject, with colors indicating diagnostic group. The red dashed horizontal line marks a corrected BAD of 0, where predicted brain age matches chronological age after bias correction. Positive corrected BAD values indicate an older-appearing brain relative to chronological age, whereas negative values indicate a younger-appearing brain. Substantial overlap is observed among the three groups, although variability in corrected BAD is present across the full age range.

ACKNOWLEDGMENT

This work was supported by National Institutes of Health training grant T32 AG082658 from the National Institute on Aging.

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